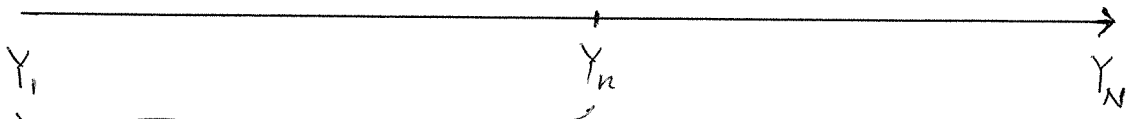
The general setup



based on these n observations to build a time series model

based on this model we can find these estimated values

$\hat{Y}_1 \cdots \hat{Y}_n \; \hat{Y}_{n+1} \cdots \hat{Y}_N \; \hat{Y}_{N+1} \cdots$

then extend the model and we get the forecast from (n+1) to N, this period is called ex post period.

Provided that the forecast error is not large, then we can extend further more based upon the built model.

further extend the forecast period due to the forecast error is small. This period of forecast is called ex ante period.

Measuring the forecast error

$$e_t = Y_t - \hat{Y}_t$$

1. The mean absolute error

$$MAE = \frac{\sum_{t=1}^{n} |e_t|}{n}$$

2. The mean of absolute percentage error

$$MAPE = \frac{\sum_{t=1}^{n} \frac{|e_t|}{Y_t}}{n}$$

3. The mean squared error

$$MSE = \frac{\sum_{t=1}^{n} e_t^2}{n}$$

this is one of the measure of MSE (biased) for time series measurement of error.

$$MSE = \frac{\sum_{t=1}^{n} e_t^2}{n-2}$$

this is an unbiased measure of MSE for statistical analysis.

N.B. Data : (time, observed value)

$(1, Y_1), (2, Y_2), \ldots, (n, Y_n)$

n pairs of data

$\therefore$ we have n degrees of freedom

We build the linear trend model

$$\hat{Y}_t = \hat{\alpha} + \hat{\beta} t, \quad t = 1, 2, \ldots, n.$$

$Var(\varepsilon_t) = Var(Y_t - \hat{Y}_t)$ where $\hat{Y}_t$ depends on $\bar{t}$ and $\bar{Y}_t$

$n\bar{t} = \sum_{i=1}^{n} t_i$ and $n\bar{Y}_t = \sum_{i=1}^{n} Y_i$

are 2 restrictions, hence lost 2 degrees of freedom from n.

4. The root mean squared error

$$RMSE = \sqrt{MSE} = \sqrt{\frac{\sum_{t=1}^{n} e_t^2}{n}}$$

5. Theil's U

$$U = \frac{\sqrt{\frac{1}{n}\sum_{t=1}^{n}(Y_t - \hat{Y}_t)^2}}{\sqrt{\frac{1}{n}\sum_{t=1}^{n}\hat{Y}_t^2} + \sqrt{\frac{1}{n}\sum_{t=1}^{n}Y_t^2}}$$

Theil's U statistic is a relative accuracy measure (no units) that compares the forecasted results with the results of forecasting with minimal historical data. It is clear that U has a value between 0 and 1. A small value of U indicates a good forecasting model. Since the numerator is based on $\sum_{t=1}^{n}(Y_t - \hat{Y}_t)^2 = \sum_{t=1}^{n} e_t^2$. If $\sum_{t=1}^{n} e_t^2$ is small, then U will be small (i.e., close to zero).

Some simple forecast models

1. The naive forecast

$$\hat{Y}_{t+1} = Y_t$$

which takes the previous value (Y_t) as the forecast of the next period ($\hat{Y}_{t+1}$).

2. Simple moving average

k-point moving average forecast

$$= \hat{Y}_{t+1} = \frac{\sum_{i=0}^{k-1} Y_{t-i}}{k}$$

$$= \frac{Y_t + Y_{t-1} + Y_{t-2} + \cdots + Y_{t-(k-1)}}{k}$$

$Y_{t-(k-1)}$
$\vdots$
Y_{t-2}
Y_{t-1}
Y_t

which takes the previous k observations and then average it.

3. Linear trend forecast (depends on time t)

$Y_t = \alpha + \beta t + \varepsilon_t$ where ε_t has mean 0 and variance σ^2.

(model)

$\hat{Y}_t = \hat{\alpha} + \hat{\beta} t$

$\overset{or}{=} a + bt$

(estimated model)

This is a linear regression model based on time t.

$\hat{\beta}$ estimated slope

$\hat{\alpha}$ estimated y-intercept

[We will go through linear regression later.]

4. Causal regression model (depends on explanatory or independent variable X)

$Y_t = \alpha + \beta X_t + \varepsilon_t$ where ε_t is the random error with mean 0 and variance σ^2.

$\hat{Y}_t = \hat{\alpha} + \hat{\beta} X_t$

$\overset{or}{=} a + b X_t$

Example: Annual revenues of a certain company p3

Year	Y_t Revenue	t	Naive	3-period MA	Trend $Y_t = 374.41 + 42.89t$
00	375.2	0			
01	414.2	1			
02	458.5	2			
03	511.9	3			
04	542.0	4			
05	588.1	5			
06	656.4	6	588.1	① 547.3	631.75 ≃ 631.8
07	722.5	7	656.4	② 595.5	674.64 ≃ 674.6
08	804.2	8	722.5	③ 655.7	717.53 ≃ 717.5
09	896.8	9	804.2	④ 727.7	760.42 ≃ 760.4
10	957.4	10	896.8	⑤ 807.8	803.31 ≃ 803.3
11	1038.7	11	957.4	⑥ 886.1	846.2
12	1069.3	12	1038.7	⑦ 964.3	889.09 ≃ 889.1
13	1167.9	13	1069.3	⑧ 1021.8	931.98 ≃ 932.0

① : $\frac{511.9 + 542 + 588.1}{3} = 547.3$

② : $\frac{542 + 588.1 + 656.4}{3} = 595.5$

③ : $\frac{588.1 + 656.4 + 722.5}{3} = 655.6$

④ : $\frac{656.4 + 722.5 + 804.2}{3} = 807.8$

⑤ : $\frac{722.5 + 804.2 + 896.8}{3} = 807.8$

⑥ : $\frac{804.2 + 896.8 + 957.4}{3} = 886.1$

⑦ : $\frac{896.8 + 957.4 + 1038.7}{3} = 964.3$

⑧ : $\frac{957.4 + 1038.7 + 1069.3}{3} = 1021.8$

$\sum_{i=0}^{5} t_i = 15 \quad \sum_{i=0}^{5} t_i^2 = 55$

$\sum_{t=0}^{5} Y_t = 2889.9$

$\sum_{t=0}^{5} t Y_t = 7975.4$

Linear Trend model

$$b = \hat{\beta} = \frac{\sum_{t=0}^{5}(t-\bar{t})(Y_t-\bar{Y})}{\sum_{t=0}^{5}(t-\bar{t})^2} = \frac{\sum_{t=0}^{5} t Y_t - \frac{\left(\sum_{t=0}^{5} t\right)\left(\sum_{t=0}^{5} Y_t\right)}{6}}{\sum_{t=0}^{5} t^2 - \frac{\left(\sum_{t=0}^{5} t\right)^2}{6}}$$

$$= \frac{7975.4 - \frac{15(2889.9)}{6}}{55 - \frac{(15)^2}{6}} = \frac{750.65}{17.5} = 42.8942$$

$$a = \hat{\alpha} = \bar{Y} - \hat{\beta}\bar{t} = \frac{2889.9}{6} - 42.8942\left(\frac{15}{6}\right)$$

$$= 374.4145 \simeq 374.41$$

1.4 Simple exponential smoothing (for constant process)

Let $X_t = b + \varepsilon_t$, where X_t takes on the observations $x_1, x_2, \ldots, x_T$. The setup of the simple exponential smoothing (SES) method is:

$$\hat{b}(T) = \hat{b}(T-1) + \alpha[x_T - \hat{b}(T-1)], \text{ where } 0 < \alpha < 1.$$

Rationale: we modify the old estimate (forecast) by some fraction of the forecast error.

Note that $\hat{b}(T-1)$ is the estimate of $\hat{b}(T)$, α is the fraction and $x_T - \hat{b}(T-1)$ is the forecast error.

Now, let $S_T \equiv \hat{b}(T)$, then we have

$$S_T = S_{T-1} + \alpha[x_T - S_{T-1}] = S_{T-1} + \alpha x_T - \alpha S_{T-1}$$
$$= \alpha x_T + (1-\alpha) S_{T-1}$$

Note that S_{T-1} is the estimate of S_T.

Furthermore, for $X_t = b + \varepsilon_t$, thus

$$X_t - b = \varepsilon_t$$

then $$E(X_t - b) = E(\varepsilon_t) = 0$$

$\therefore$
$$E(X_t) = b$$
$$= \hat{b}(T)$$
$$= S_T$$

1.4.1 Properties of simple exponential smoothing forecast

We have

$$S_T = \alpha x_T + (1-\alpha) S_{T-1}$$
$$S_{T-1} = \alpha x_{T-1} + (1-\alpha) S_{T-2}$$
$$S_{T-2} = \alpha x_{T-2} + (1-\alpha) S_{T-3}$$
$$\vdots$$
$$S_1 = \alpha x_1 + (1-\alpha) S_0$$

Now, by substitution, we have

$$S_T = \alpha x_T + (1-\alpha)\left[\alpha x_{T-1} + (1-\alpha) S_{T-2}\right]$$
$$= \alpha x_T + \alpha(1-\alpha) x_{T-1} + (1-\alpha)^2 S_{T-2}$$
$$= \alpha x_T + \alpha(1-\alpha) x_{T-1} + \alpha(1-\alpha)^2 x_{T-2} + (1-\alpha)^3 S_{T-3}$$
$$\vdots \quad \text{keeping onwards}$$
$$S_T = \alpha \sum_{k=0}^{T-1} (1-\alpha)^k x_{T-k} + (1-\alpha)^T S_0$$

Thus, S_T is a weighted average of all past observations.

Moreover,

$$\lim_{T\to\infty} (1-\alpha)^T S_0 = 0$$

$$E(S_T) = E\left[\alpha \sum_{k=0}^{\infty} (1-\alpha)^k x_{T-k}\right] = \alpha \sum_{k=0}^{\infty} (1-\alpha)^k E(x_{T-k})$$
$$= E(x_{T-k}) = b.$$

$\because \sum_{k=0}^{\infty} (1-\alpha)^k = 1 + (1-\alpha) + (1-\alpha)^2 + \cdots = \frac{1}{1-(1-\alpha)} = \frac{1}{\alpha}$

$\therefore \alpha \sum_{k=0}^{\infty} (1-\alpha)^k E(x_{T-k}) = \alpha \times \frac{1}{\alpha} \times E(x_{T-k}) = b$

Furthermore, $E(S_T) = b$

$E(S_{T-1}) = b$

$E(S_{T-2}) = b$

$\vdots$

$E(S_{T-k}) = b$

$\therefore$ S_T is an unbiased estimate of b, thus it is natural to have

$$\hat{x}_{T+\tau} = S_T$$

N.B. 1. We frequently use the average of the recent observations as S_0.

2. In terms of updating, exponential smoothing is a very convenient procedure.

3. For $S_T = \alpha \sum_{k=0}^{T-1} (1-\alpha)^k S_{T-k} + (1-\alpha)^T S_0$, it is exponentially decay with α, where $0 < \alpha < 1$. That is, why the S.E.S. method is exponentially smoothing, in fact, it is exponentially decaying.

Example : Simple Exponential Smoothing

Monthly sales of TV in the past 6 months

$S_0 = 393.13 \quad \alpha = 0.1$ N.B. S_0 is the forecast of x_1 (X_1) $393.13 \simeq 393$

the smoothed statistics and the one-step forecasts

The model is

$$S_T = \alpha x_T + (1-\alpha) S_{T-1}$$

$$S_1 = 0.1 x_1 + 0.9 S_0$$

$$= 0.1(330) + 0.9(393.13) = 386.817 \simeq 386.82$$

It will be the forecast of x_2 (or X_2) $= 386.82 \simeq 387$

$$S_2 = 0.1 x_2 + 0.9 S_1 = 0.1(410) + 0.9(386.82) = 389.138$$

$$\simeq 389.14 \Longrightarrow X_3 = 389$$

$$S_3 = 0.1 x_3 + 0.9 S_2 = 0.1(408) + 0.9(389.138) = 391.0242$$

$$\simeq 391.02 \Longrightarrow X_4 = 391$$

$$S_4 = 0.1 x_4 + 0.9 S_3 = 0.1(514) + 0.9(391.0242) = 403.3218$$

$$\simeq 403.32 \Rightarrow X_5 = 403$$

$$S_5 = 0.1 x_5 + 0.9 S_4 = 0.1(402) + 0.9(403.3218) = 403.1896$$

$$\simeq 403.19 \Rightarrow X_6 = 403$$

$$S_6 = 0.1 x_6 + 0.9 S_5 = 0.1(343) + 0.9(403.1896) = 397.1706$$

$$\simeq 397.17 \Rightarrow X_7 = 397$$ (This value does not show in table.)